

Destined to Receive: The Impact of Remittances on Household Decisions in Armenia

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Abstract

The paper uses data from Armenia to test the implications of remittance flows on behavior of receiving households. We find that remittance-receiving households work fewer hours and spend less on the education of their children. While saving more, these households are not leveraging their savings to borrow from the banking system to expand their business activities. This evidence suggests that the benefits of remittances might be overstated and emphasizes the importance of measuring their impact in a general- rather than a partial-equilibrium context.

1. Introduction

The New Economics of Labor Migration (NELM) offers a rich agenda for theoretical and empirical research with potential implications going far beyond the determinants of migration and remittances. One aspect of the NELM agenda that has received little attention deals with the implications of migration-driven remittances on the micro-economic incentives of family members left behind. This is an important issue to tackle if one is to fully grasp the developmental impact of remittances on receiving countries.

The worldwide flow of remittances has grown tremendously in recent years, becoming the second source of development finance after foreign direct investment (World Bank, 2005). While a general consensus exists on the poverty-reducing implications of remittances, if looked through the prism of their overall developmental impact in a general-equilibrium setting, remittances could be a blessing in disguise. Apart from the macroeconomic policy challenges of managing remittance flows, there exist other factors that may swing the pendulum in the other direction. Often overlooked in this regard are the loss of productive labor for the home country as well as the potential disincentives to work, study, and stay in the home country for the family members left behind.¹ Empirical evidence presented in this paper suggests that the benefits of migration and remittances might indeed be overstated. Our estimates of the effect of remittances on labor supply, education spending, and saving and borrowing behavior help shed more light on the impact of remittances on future growth and development prospects.

This paper examines the microeconomic effects of remittances using data from Armenia, one of the most industrialized and socially (i.e. in terms of education and health) advanced republics of the former Soviet Union. Remittances in Armenia, which may have reached 20% of GDP in recent years (CBA, 2006), have certain peculiarities at the micro level. Relatively strong (extended) family ties observed in Armenia

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(Alesina and Giuliano, 2007) may have increased the demand for remittances, resulting in transfers going to households of which the migrant is not an immediate member.² Remittances in Armenia also originate from a large diaspora, the members of which do not necessarily have immediate family in Armenia but remit money to their distant relatives or friends. Thus, we observe a sizable amount of (out-of-household) transfers not directly linked with migration.

We present empirical evidence on the microeconomic effects of remittances using data from a region that has not been previously studied. The results are quite intriguing: we find that remittances reduce incentives to work and lead to less spending on education. In addition, we find evidence that remittance-receiving families save more. This, however, does not result in more borrowing from the banking system, contrary to what the remittances-as-collateral argument would suggest.

2. Review of the Empirical Literature on the Impacts of Remittances

The focus of the recent remittances literature has shifted away from the determinants of remittances toward the (macro and micro) implications of the remittance flows. Although most of the empirical work has focused on migrant-exporting countries with rather similar characteristics, the debate about the impact of remittances is far from over.

Using cross-country macro data, Chami et al. (2005) report a negative effect of remittances on growth and productivity. Another macro study, Acosta et al. (2007), finds that in addition to the usual nominal exchange rate channel, remittances result in a shrinkage of, and resource reallocations away from, the tradable sector and thereby increasing the production costs of the otherwise labor-intensive nontradable sector. Using micro data from Morocco, van Dalen et al. (2005) find that remittances have a potential to stimulate further migration among the family members left behind. These studies all point out that in the presence of these effects, the benefits of remittances, if any, could be less pronounced. In contrast, using panel data for countries in Latin America and the Caribbean, Mundaca (2009) finds that remittances can have significant positive long-run growth effects. Lokshin et al. (2010) estimate that almost 20% of the decline in poverty in Nepal between 1995 and 2004 is due to increased work-related migration and remittance inflows.

In terms of the microeconomic behavioral changes driven by remittances, the literature has focused on the impact of remittances on household expenditure (including such subcomponents as expenditures on investment goods, health, and education) and labor–leisure allocation. While most studies find that remittances boost consumption, results vary as to how much of the remittance flows get siphoned away for more productive uses. Studies that find a positive relationship between remittances and various types of household investments include Taylor (1992), Adams (1998), Rozelle et al. (1999), Dustmann and Kirchkamp (2002), Woodruff and Zenteno (2007), and Yang (2008). In contrast, others have argued that remittances are absorbed into immediate consumption and rarely finance productive investments (Brown and Ahlburg, 1999).

As for the impact of remittances on education (i.e. expenditures, attendance, etc.) here too the empirical evidence is mixed. Cox Edwards and Ureta (2003), Hanson and Woodruff (2003), Lopez Cordoba (2004), and Yang (2008) find a positive relationship between education and remittances. In contrast to these studies, McKenzie (2005) and Acosta (2006) find that remittances have a negative impact on education. Finally, a

number of papers also examine the effect of remittances on labor supply and participation decisions (Acosta, 2006; Amuedo-Dorantes and Pozo, 2006; Yang, 2008). These results are also mixed.

3. Empirical Estimation

Background and Data Description

Armenia is an interesting, if not extreme, case for studying migration and remittance behavior. A highly industrialized and well-educated society³ at the dusk of the Soviet Union, Armenia underwent severe economic contractions imposed by the collapse of trade links (following the fall of the Soviet Union) and a full-scale conflict with neighboring Azerbaijan over Nagorno Karabakh during 1991–94.

Although some out-migration took place during the conflict itself, an exodus of the population did not begin until it became clear that the government had not delivered on its promises of economic reform after the war. While GDP grew slightly in 1994, after falling by almost half in 1992–93, and inflation was declining from its triple-digit war levels, social conditions were still severe. Corruption—which began to take root before the war—was spreading rapidly and taking new forms. This was too much to bear for a population that had uniformly endured the hardships of war:⁴ given the popular support for the war, hardships during it were easier to justify than the hardships that followed. The traditional Armenian diaspora in Western countries and Russia was rapidly “reinforced” by members of the new diaspora, causing Armenia to become one of the top population-exporting countries in the world (measured as a percentage of pre-emigration population).⁵ Remittances became a key source of income for many household members and related families left behind in Armenia.⁶ While generally assumed to be sizable—a sensible assumption in an environment with low wage rates—the underlying incentive effect of remittances on receiving families has not been subject to much research. The question of whether remittances have provided incentives to migrate for those left in Armenia has also not been researched.

In addition to these incentive effects, the underlying migration resulted in a loss of productive labor for the economy. A study by the International Organization of Migration (IOM, 2002) notes that the share of migrants with higher education is double that of the national average, suggesting a drain on average economy-wide knowledge and skills. While the actual loss of productivity from migration and the brain drain lies beyond the scope of this study, this nevertheless is an important factor to keep in mind in the Armenian context.⁷

For our analysis, we used the 2004 *Integrated Living Standards Measurement Survey* conducted by the National Statistical Service of Armenia. This is a comprehensive survey covering a wide range of household activities, including migration and remittances. The dataset is nationally representative, containing data for over 28,000 individuals from over 6800 households.

The dataset reveals peculiarities of remittance flows in the Armenian context. As conjectured above, strong family ties are likely to be the factor behind the remittances going to households of which the migrant is not an immediate member. In addition, remittances in Armenia are also likely to originate from the traditional/old diaspora, members of which remit money to their distant relatives or friends, and not necessarily to immediate family members in Armenia. Indeed, 23% of all surveyed households report receiving transfers from abroad, but only 14% report having a migrant abroad. Only 37% of all remittances are migration-related (i.e. 63% comes from out-of-

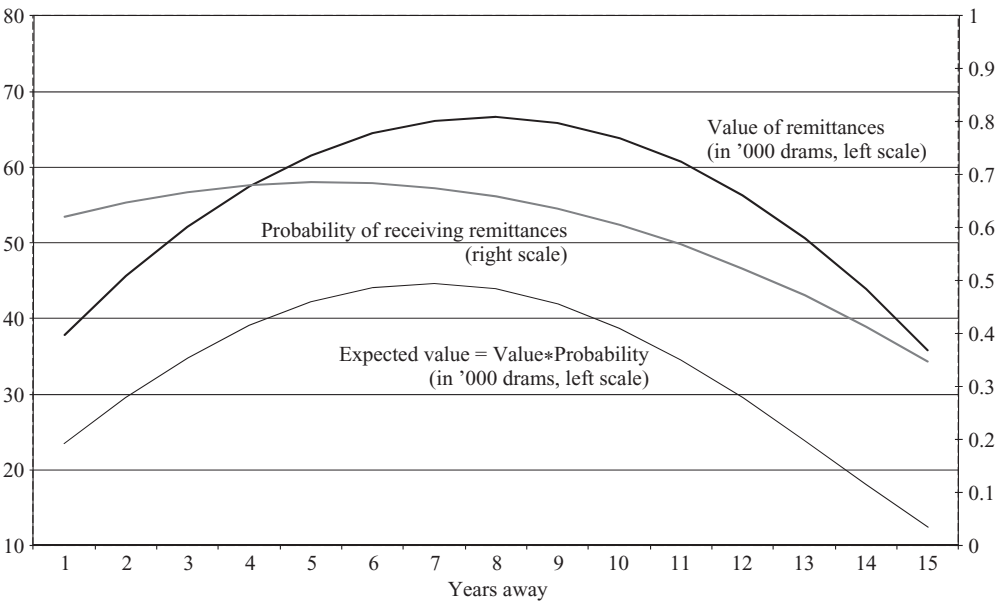


Figure 1. Impact of Time Away on Remittance Behavior

Notes: Based on the following regressions: Probability = $0.59 + 0.04 * (\text{Time away}) - 0.004 * (\text{Time away})^2$ and Value = $28.9 + 9.57 * (\text{Time away}) - 0.61 * (\text{Time away})^2$ using only households with migrants. Estimates of other regressors (not reported above) are included in the constant terms at their mean values.

household sources). In terms of their impact on the behavior of the recipients, however, transfers from nonmigration-related sources are likely to have similar implications as those related to migration, and so we did not exclude them from the dataset.

One of the peculiarities of the remittance pattern in Armenia is that migration-driven remittances are likely to change with the migrant's time away. Figure 1 depicts the relationship between time away and (the probability as well as the amount of) remittances received by households. The results show that both the propensity to remit and the amounts remitted decline as time passes—after reaching their respective maxima at five and seven years of migration—and are consistent with the remittance decay hypothesis.

In addition to the existence of the above peculiarities, it is worth noting that migration and remittance indicators may be subject to measurement error. Given that the survey has been conducted by an official agency, households with migrants may underreport remittances to understate their true income. Indeed, only 62% of families with migrant members report receiving remittances: while any number below 100 would be consistent with the migration decay hypothesis, this number is likely to be on the low side, given the strong family ties in Armenia. In addition, remittance-receiving households may choose to underreport the number (or existence) of migrants in the household, if the latter left the country, say, to avoid compulsory military service.

While indeed comprehensive, the dataset allows us to test only a limited subset of the recipients' choices affected by remittances. We attempted to test some of the most important relationships, leaving the rest to be tested in the future. With this caveat, we move on to present the econometric model and the estimation results.

Baseline Regression

Structural model The key issue to tackle when estimating the impact of remittances on economic outcomes is the potential simultaneity of remittances/migration and the economic outcomes in question. Because of certain (unobserved) characteristics, families who send members abroad (and then receive remittances) could also be those who choose to consume more leisure. Similarly, the presence of credit constraints that may prevent families from financing a member's trip abroad, may also limit the amount of healthcare spending they can afford. Unless one controls for those family characteristics and the variables that may explain the credit constraints, one's estimates of the impact of remittances on leisure and healthcare spending, respectively, would be biased.

To properly capture the motives to migrate and separate the impact of migration from that of remittances, we follow Rozelle et al. (1999) by employing a three-stage structural model to study the behavior of households. In the first stage, an individual's decision to migrate, M , is determined as follows:

$$M = \alpha_1 + \beta_1 \cdot HH + \gamma_1 \cdot Z_M + \varepsilon, \quad (1)$$

where HH is the set of observed household characteristics, and Z_M is a vector of (excluded) instruments to help properly identify M .

In the second stage, given the individual's decision to migrate, the decision to remit, R , will be determined as:

$$R = \alpha_2 + \beta_2 \cdot HH + \gamma_2 \cdot Z_R + \theta_2 \cdot \bar{M} + \sigma, \quad (2)$$

where $\bar{M}$ is the fitted value of the migration variable from equation (1), and Z_R is a vector of (excluded) instruments to properly identify R .⁸

Finally, in the third stage, the microeconomic outcomes will be determined as a function of remittances in the following way:

$$Outcome = \alpha_3 + \beta_3 \cdot HH + \gamma_3 \cdot \bar{R} + \theta_3 \cdot \bar{M} + \omega, \quad (3)$$

where $\bar{R}$ is the fitted value of the remittance variable from equation (2). Given their developmental impact (and the variables available in the dataset), we choose the following indicators as outcomes: (i) per capita hours worked by the members of the household, (ii) per student expenditures on education, (iii) household savings, and (iv) a dummy variable indicating whether the household members had borrowed from the banking sector during the 12 months preceding the survey. The dummy variable is used to test whether remittances allow households to overcome borrowing constraints by acting as collateral.⁹

It is important to note that the error terms in equations (1) and (2) are likely to be correlated because of, for example, an unobservable factor that could influence both the propensity to migrate and to remit, once the individual has migrated. Therefore, joint estimation of the model equations is preferred (to the stage-by-stage estimation) on the grounds that it will be asymptotically more efficient (Hayashi, 2000, p. 273). We, therefore, applied the three-stage least squares (3SLS) technique to estimate the full model.

Instruments and control variables The choice of instruments for migration and remittances is critical for properly identifying these variables and avoiding problems of

estimation associated with inadequate instruments. We would, therefore, like to discuss our thinking behind the selection of instruments in some detail.

Traditional models of migration behavior may not offer sufficient insight into the choice of meaningful instruments for migration. As Bauer and Zimmerman (1999) note, wage-differential-based economic models fail to adequately explain the variations in migration across countries. Instead, a study by the World Bank (2006) emphasizes the importance of broader quality-of-life considerations and political factors in explaining migration. The study goes on to say that “differences in political stability, human rights situations, and the general rule of law may also affect migration, because these factors serve as proxies for the level of individually perceived insecurity.”¹⁰ With this in mind, we included a measure of corruption to proxy for differences in push factors behind migration across regions. Obtained via a nationwide survey conducted in 2000, the variable measures the share of respondents in every region who mentioned corruption and absence of law and order as obstacles for rising living standards (see the Appendix for the source).

In addition, to proxy for pull factors, we used historic regionwide migration rates to capture the ease of migration via the presence of migration networks abroad. This approach is consistent with those used by Rozelle et al. (1999), Hanson and Woodruff (2003), McKenzie (2005), and Woodruff and Zenteno (2007), who use the migration pattern in the migrant’s area of origin as an instrument for migration.

The choice of instruments for remittances was guided by a similar logic. Following Amuedo-Dorantes and Pozo (2006), who used the unemployment rate in the region to instrument remittances, we used the regionwide unemployment rate among men as an instrument. In addition, to capture some of the motives behind out-of-household (and to some extent also the migration-driven) remittances, we used the ratio of vulnerable population to the total regional population as an instrument to measure the demand for remittances. To give credibility to these instruments, we formally test for the presence of correlations between the instruments and the dependent variables in equation (3) above.¹¹

Interestingly enough, all four region-level indicators provide enough variance across regions to enable statistical testing and to produce meaningful results. Identification of migration and remittance indicators comes from the correlation between these indicators and the region-level instruments taken separately and in interaction with the household’s average level of education (percentage of household members with postsecondary degrees).¹²

The following household-level control variables are included in the regressions: (i) age of the household head (and its squared value); (ii) average age of working-age members of the household (and its squared value); (iii) dummy for male household head; (iv) percentage of household members with postsecondary education; (v) percentage of household members above 64; (vi) percentage of household members under 6; (vii) percentage of women in the household; (viii) size of the household; and (ix) dummy variables to distinguish the rural and noncapital urban areas from the capital. All variables and their sources are summarized in the Appendix.

To capture the impact of household wealth/assets on (the probability of receiving) remittances and subsequent microeconomic behavior, we also included a set of dummy variables based on a self-assessment of dwelling conditions.¹³ A positive relationship between wealth and remittances here would be consistent with self-interest motives, while a negative relationship would be indicative of the presence of altruistic motives. Similarly, we expect this measure of wealth to also have an impact on outcome indicators.

Summary of results—first and second stage The results of the estimation of equations (1) and (2) are presented in Table 1. Some patterns of migration behavior are noteworthy. As predicted, the presence of migration networks encourages migration: the coefficient on past migration from the region is positive and statistically highly significant. However, corruption and rule-of-law-related problems are associated with a lower probability of migration. This is possible either because corruption reduces the ability of households to meet the costs of migration (other things being equal, residents of regions with more corruption cannot afford to migrate) or because potential migrants fear leaving behind family members in a corruption-infested environment. It is likely that in cases where corruption is seen as a serious enough threat for the future wellbeing of the family, the entire family would relocate. Unfortunately, our dataset does not contain information on this category of households, and does not allow us to make a more complete inference regarding this issue.

Age of both the head and the working-age members of the household are associated with more migration. Surprisingly, households with male heads are less likely to have a migrant in the family. Consistent with most of the literature, households with more children and women are less likely to have a migrant while larger households are more likely to have a migrant. The impact of wealth on migration has a peculiar inverted U-shaped relationship. Households with average wealth (proxied by “good” and “normal” quality of housing) are more likely to have a migrant than the two extremes (i.e. those with “excellent” and “not good” quality of housing). This is consistent with Hancilova and Minasyan (2005), who, in the Armenian context, argue that the poor cannot afford to send migrants abroad, while the rich do not have the incentives to do so. Finally, it appears that households from urban areas outside of the capital are the most likely ones to have a migrant, followed by rural households, and households in the capital city. In the Armenian context this reflects the pattern of availability of economic opportunities across these areas, with Yerevan having more to offer, followed by rural areas (with opportunities to engage in agricultural activities), leaving the urban dwellers outside of the capital with little private or public sector activity to live off.

The remittances equation too has some interesting insights to offer. As expected, the coefficient on migration is positive and significant: the magnitude of the coefficient (61%) is very close to that observed in the (uncontrolled) sample (62%). Both high unemployment and share of vulnerable population discourage remittances, providing strong support for the self-interest-driven nature of these flows. In addition, transfers to more mature families are both smaller in size and less likely than to their younger counterparts. While it is plausible that the per capita remittance flow declines with household size, it is less clear why the probability of receiving remittances would be lower in larger households. It may, however, be because larger households are perceived (by both the migrants as well as the out-of-household remitters) as more secure/less vulnerable, requiring less to sustain their consumption and livelihood. It could also be that the bequest-related motives are weakened as the migrant faces competition from other members of the household for an inheritance. Finally, larger households may have more opportunities to share risks. Therefore, to the extent that remittances are (self-enforceable) contracts to share risks, larger households will be less likely to receive remittances. All in all, the existence of this relationship provides additional support for the self-interest motive for remittances.

In terms of the age composition of the households, it is interesting to note that while families with more elderly people are just as likely to receive transfers as others, they receive smaller amounts, which most likely reflects their needs (which are typically more modest, compared to those of younger household members). In terms of the

Table 1. Results of the First- and Second-Stage Regressions

	Migration		Remittances (yes/no)		Remittances ('000 drams)	
	Coefficient	Standard error	Coefficient	Standard error	Coefficient	Standard error
Migration			0.611	[0.189]***	100.628	[30.313]***
<i>Instruments for M</i>						
Corruption	-0.002	[0.001]**				
Emigration	0.012	[0.002]***				
Corruption * Ppostsec	-0.002	[0.003]				
Emigration * Ppostsec	0.005	[0.009]				
<i>Instruments for R</i>						
Unemploy			-0.352	[0.136]***	-62.666	[21.909]**
Vulnerable			-0.001	[0.001]	-0.454	[0.151]***
Unemploy * Ppostsec			-0.361	[0.440]	8.006	[70.156]
Vulnerable * Ppostsec			-0.003	[0.003]	-0.143	[0.437]
<i>Control variables</i>						
Age	0.010	[0.002]***	-0.002	[0.003]	-0.591	[0.480]
Age-sq	0.000	[0.000]***	0.000	[0.000]	0.004	[0.004]
Aveagework	0.004	[0.002]**	-0.006	[0.002]***	-1.558	[0.332]**
Aveagework-sq	0.000	[0.000]***	0.000	[0.000]***	0.028	[0.005]***
HHheadmale	-0.019	[0.011]*	-0.008	[0.013]	-0.653	[2.135]
Ppostsec	-0.024	[0.093]	0.220	[0.213]	8.188	[34.069]
Pelderly	-0.024	[0.031]	-0.035	[0.038]	-11.565	[6.138]*
Pkids	-0.203	[0.039]***	0.003	[0.049]	8.723	[7.941]
Pwomen	-0.128	[0.022]***	0.051	[0.029]*	12.212	[4.776]**
Hhsize	0.024	[0.003]***	-0.009	[0.005]*	-4.797	[0.743]***
Timeaway			-0.011	[0.027]	-1.704	[4.309]
Good	0.047	[0.018]**	0.105	[0.022]***	12.369	[3.579]***
Normal	0.026	[0.015]*	0.080	[0.018]***	4.856	[2.849]*
Not good	0.012	[0.016]	0.037	[0.019]**	1.575	[3.017]
Urban (noncapital)	0.067	[0.010]***	-0.010	[0.018]	-9.029	[2.975]***
Rural	0.023	[0.012]*	-0.055	[0.019]***	-11.222	[3.076]***
Constant	-0.363	[0.072]***	0.427	[0.115]***	91.86	[18.532]***
No. of observations	6816		6816		6816	

Notes: Standard errors in square brackets. ***, **, and * denote significance at 1%, 5%, and 10% confidence levels, respectively.

gender composition, households with a greater share of women are more likely to receive transfers (and in larger amounts). This could be the result of a higher demand imposed on families by costs associated with the presence of women in families, such as wedding/marriage- or health-related expenditures. Regarding the effect of receiving a family's wealth on remittances, there exists a similar inverted U-shaped relationship between the two. Both the probability and the amounts received are greater in households with average housing quality (i.e. "good" and "normal" categories), compared to households with "not good" and "excellent" housing. Outside of the households with "excellent" housing (representing 9% of sample households), this result provides further evidence in support of the self-interest motive in remittance behavior. Finally, it is interesting to note that households from noncapital urban and rural areas are less likely to receive remittances than people from the capital. However, if viewed in conjunction with the finding that they are also more likely to migrate, this suggests that migrants originally from outside of the capital earn less abroad (and, therefore, remit less *ceteris paribus*) or that they are less likely to return (hence no need to remit as much).

Summary of results—third stage Prior to discussing the results of equation (3), a few words in support of the choice of instruments in equation (3) are in order. It turns out that the Anderson–Rubin overidentification test could not be rejected at any reasonable level in any of the four third-stage equations reported in Table 2, suggesting that the instruments pass the exogeneity test. Similarly, Hansen's J-statistic is not significant at any level for three out of the four regressions providing sufficient comfort to proceed.¹⁴ We therefore discuss the results reported in Table 2 for each of the four dependent variables.

The impact of remittances on hours supplied to the market (column (2)) is as predicted by the model. The coefficient is negative and significant and its magnitude is rather large. Indirectly, this serves a blow to the hypothesis that remittances help relax budget constraints for families staying behind and help expand their production possibilities frontier—to the extent that capital and labor are complements in the production process, it is not clear why households that invest more (due to remittances) would work fewer hours. Interestingly enough, while insignificant, the coefficient on migration is positive, suggesting perhaps that the absence of a family member tightens the family's discipline, making the remaining household members work harder to attain the same levels of utility and welfare. The magnitude of the coefficient is not large enough, however, to offset the disincentive effect of remittances.

Other findings of this regression are also worth flagging: (i) it appears that women in Armenia supply as many hours to the market as men, if not more; (ii) the presence of both elderly and children in families result in more hours supplied to the market (even though they themselves do not); and (iii) the residents of rural households on average work 1.25 hours more than those in urban locations. Finally, it is interesting to note that even after controlling for the share of elderly and children in the household (because their contribution is zero), larger households supply fewer aggregate hours to the market than smaller households. This supports our earlier conjecture that family size could provide additional benefits (such as insurance against volatility and vulnerability) and, therefore, lead to fewer hours worked.

The impact of remittances on education spending (column (3), Table 2) is perhaps the most controversial of our findings. The negative (and significant) coefficient here could be indicative of two things. First, it is possible that members of remittance-receiving households are likely to later migrate themselves and, therefore, not value the

Table 2. Results of the Third-Stage Regressions

	<i>Hours worked</i>	<i>Education expenditures</i>	<i>Savings</i>	<i>Banking</i>
Remittances (hat)	-7.677 [3.250]**	-355.958 [70.834]***	213.0 [85.889]**	0.294 [0.552]
Migration (hat)	1.329 [1.527]	215.365 [37.330]***	-95.527 [40.357]**	-0.253 [0.259]
Age	-0.062 [0.024]***	1.125 [0.955]	-0.017 [0.514]	0.006 [0.003]**
Age-sq	0.001 [0.000]***	-0.008 [0.009]	0.002 [0.005]	0.000 [0.000]**
Aveagework	0.374 [0.038]***	-2.703 [1.780]	1.306 [0.613]**	0.005 [0.004]
Aveagework-sq	-0.004 [0.000]***	0.023 [0.024]	-0.024 [0.010]**	0.000 [0.000]
HHheadmale	0.190 [0.111]*	3.466 [4.631]	4.576 [2.736]*	0.020 [0.015]
Ppostsec	1.531 [0.153]***	111.729 [7.966]***	3.811 [4.113]	-0.002 [0.023]
Pelderly	1.375 [0.335]***	-5.275 [20.611]	4.198 [8.117]	-0.056 [0.046]
Pkids	0.782 [0.355]**	-61.927 [22.633]***	17.152 [9.229]*	-0.109 [0.050]**
Pwomen	0.081 [0.288]	49.867 [11.159]**	-7.81 [6.109]	-0.023 [0.035]
Hhsize	-0.131 [0.035]***	-9.019 [1.497]***	0.185 [0.879]	0.022 [0.005]***
Good	1.527 [0.408]***	52.502 [10.847]***	-13.396 [10.271]	-0.129 [0.064]**
Normal	1.167 [0.321]***	32.75 [8.207]***	-16.043 [7.843]**	-0.082 [0.049]*
Not good	0.558 [0.224]**	14.897 [7.114]**	-7.967 [4.919]	-0.017 [0.029]
Urban (noncapital)	0.213 [0.175]	-12.842 [6.481]**	-10.607 [4.958]**	0.107 [0.026]***
Rural	1.275 [0.228]***	-32.408 [8.136]***	3.787 [4.562]	0.087 [0.042]**
Constant	-2.514 [1.091]**	107.37 [38.476]***	-42.715 [25.725]*	-0.061 [0.156]
Number of observations	6164	3603	6816	6816
Anderson–Rubin OR test, $\chi^2(2)$	0.720, $p = 0.70$	1.456, $p = 0.48$	2.104, $p = 0.35$	2.104, $p = 0.35$
Hansen's J-test of OR, $\chi^2(1)$	1.292, $p = 0.26$	0.272, $p = 0.60$	0.317, $p = 0.57$	34.40, $p = 0.00$

Notes: Standard errors in square brackets. ***, **, and * denote significance at 1%, 5%, and 10% confidence levels, respectively. Regional dummies are included in the regressions (not shown). Anderson–Rubin and Hansen's J-overidentification tests both test the relevance of instruments in equation (2) only.

local education as much. Second, because their consumption patterns might be under scrutiny by the remitter, the receiving households may adjust their consumption pattern to look more conservative and be centered around necessities (such as food and public services/utilities, and presumably not education and other types of spending that could be considered unnecessary from the remitter's point of view). To the extent that remittances represent a large share of the receiving family's income, for the same level of disposable income, this tendency to "simplify" the spending pattern could in fact lead to lower spending on education (in nominal terms) out of total income.¹⁵ Interestingly enough, the absence of a family member leads to higher education spending, but not enough to offset the reduction of spending associated with remittances. We also found in this context that households with a higher ratio of members with secondary and postsecondary education and those with a larger share of women spend more on the education of their children.¹⁶

The results on the effect of remittances and migration on savings (column (4)) are as expected: families with remittances accumulate more savings. However, families with migrants report fewer savings, presumably reflecting the costs of migration (to the extent they migrated within the 12 months prior to the survey; that is, during the same period where the savings are measured). The negative effect of migration is not large enough, however, to offset the accumulation of savings due to remittances.

Finally, it appears that contrary to the remittances-as-collateral hypothesis, remittances in Armenia do not result in more borrowing from commercial banks (column (5)). Although it could be argued that remittances cover all the receiving family's needs for additional capital (on top of financing the additional consumption), this would require specific and very limiting assumptions about the household production function and/or the business environment they operate in. Ironically, it appears that the banks subtract as many points for the absence of a productive family member from the household (due to migration) as they add for having the additional collateral (due to remittances) in making their lending decisions: while not statistically significant, the coefficients on remittances and migration have very close absolute values.¹⁷ The impact of collateral on the ability to borrow is, however, evident elsewhere: consistent with one's expectations, households with "excellent" housing quality are more likely to borrow from the banking system than members of all other groups.

4. Conclusions

The paper uses data from Armenia to test the implications of remittance flows on the behavior of receiving households. We have found that remittance-receiving households in Armenia work fewer hours. They also spend less on the education of their children, perhaps expecting to migrate themselves or are confined to a subsistence-type spending pattern by a combination of the remitter's oversight and moral hazard. While saving more, these households are not leveraging their savings to borrow from the banking system to expand their business activities. This provides a rather gloomy picture of the impact of remittances at the microeconomic level in Armenia.

For an environment with rapidly rising real wages, such as Armenia's, and assuming a gradual fading of family ties, our model predicts that remittances will decline over time. We find evidence in support of the remittances decay hypothesis in our data: while rising initially, both the probability of sending and the amount transferred decline over time after reaching their peaks in five and seven years, respectively. Given that the net migration from Armenia has subsided in recent years, this may have macroeconomic

implications in the form of reduced remittance flows and, therefore, lower aggregate demand in years to come.

Because of the nature of the dataset, our analysis deals with only those migrants that have family members in the home country. Anecdotal evidence suggests that migration has dealt a critical blow to Armenia's average level of education and skills and through that also to competitiveness and aggregate output. To the extent that this is reversible, policies directed at strengthening the rule of law, improving human rights, and the business environment will help at least partially recover the losses. If these policies are improved, a combination of financial savings and skills acquired abroad will likely help in setting up private businesses and accepting public service employment back home. Attempts should also be made to engage others who migrated permanently but choose to remain involved in Armenia's development. The recent focus of some students of migration and international development institutions on turning the brain drain into a brain gain (World Bank, 2006) cannot be more relevant and timely for Armenia.

Appendix. Definitions of Variables

<i>Variable names</i>	<i>Description</i>	<i>Source</i>
Hours Worked	Per capita hours supplied to the market by the working-age members of the household. The following number of hours were assumed and aggregated across the household members: 4 for part-time, 8 for full-time, 10 for overtime, and 0 otherwise	Integrated Living Standards Measurement Survey (ILSMS)
Education Expenditures	Per student expenditures on education, in 1000 drams	ILSMS
Savings	Amount saved by the household during the 12 months preceding the survey, in 1000 drams	ILSMS
Banking	Dummy variable taking value of 1 if household members had borrowed from the banking sector during the 12 months preceding the survey, and 0 otherwise	ILSMS
Age	Age of the household head	ILSMS
Aveagework	Average age of working-age members of the household	ILSMS
HHheadmale	Dummy for male household head	ILSMS
Ppostsec	Percentage of household members with post-secondary education	ILSMS
Pelderly	Percentage of household members above 64	ILSMS
Pkids	Percentage of household members under 6	ILSMS
Pwomen	Percentage of women in the household	ILSMS
Hhsize	Number of household members	ILSMS
Timeaway	Number of years that a migrant has been away	ILSMS
Good, Normal, Not good	Dummy variables based on household members' assessment of their housing conditions. The omitted category is "Excellent"	ILSMS
Rural, Urban	Dummy variables to distinguish the rural and noncapital urban areas from the capital. The omitted category is capital Yerevan	ILSMS

Appendix. Continued

<i>Variable names</i>	<i>Description</i>	<i>Source</i>
Corruption	Share of respondents in every region who mentioned corruption and absence of law and order as obstacles for rising living standards	Yeganyan and Shahnazaryan (2001, Table 5)
Emigration	Ratio of population involved in migration as a share of total population in the region	Yeganyan and Shahnazaryan (2004, Table 2.3) ^a
Unemployment	Regionwide unemployment rate among men	Authors' calculations based on ILSMS
Vulnerable	Ratio of vulnerable population as a share of total population of the region	Jrbashian (2001, Table 1)
Nonfunctioning	Regionwide ratio of nonfunctioning enterprises to total number of enterprises	UNDP (1999, Table 2.5)

Note:

^a The original survey was contained in "An Overview of External Migration Processes in the Republic of Armenia during 1991–1998," and conducted by the National Statistical Service of Armenia jointly with the TACIS Program and Eurostat.

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Notes

1. See Goldfarb et al. (1984) for an earlier study along these lines.
2. Armenia's low divorce rate—with an average of 14.6% for 1989–2005, a fraction of that in a number of Central and Eastern European countries—could also serve as an indicator of strong family ties (UNISEF TransMONEE database).

3. The industrialized and educated nature of the society/labor force is a common feature for many transition countries of the former Soviet bloc. This is also a key difference between this group and many of the Latin American countries that are widely studied in the context of migration and remittances.
4. The uniformity of the hardship is perhaps best described by the fact that no central heating was supplied and electricity was available for only one to two hours a day across the country for much of 1992–95. Food rationing was also introduced across the country.
5. Official statistics report that 18.4% of 1989 population emigrated between 1989 and 2004. The population of Armenia was 3.2 million in 2004 (World Bank, 2006). However, the official population statistics are likely to be overstated allegedly due to national security-related concerns and the opportunities for election manipulation. (For an unofficial estimate of the Armenian population see IWPR, 2000, and United States Department of State, 2002.)
6. At the macroeconomic level, remittances were welcome by the country's authorities as they provided a boost to aggregate demand and reduced poverty, and as a result helped reduce the urgency of reforms. However, they also created policy challenges in the form of high asset prices and exchange rate appreciation.
7. It is especially relevant given that since the break-up of the Soviet Union, Armenia's budgetary spending on education has declined to among the lowest in the developing world (as a share of the overall fiscal envelope and GDP).
8. Instrumenting migration and remittances also helps tackle the measurement problems described in the previous section.
9. We do not deal with total household consumption directly because of: (1) measurement issues (i.e. too many imprecisely estimated components of the consumption basket in the dataset to serve as a basis for aggregation), and (2) the difficulties of determining which elements of consumption have what kind of "productive enhancements," such as spending on housing, which could improve property values, and thus be considered borderline investment.
10. This is consistent with Ghatak et al. (1996), who mention the importance of the domestic political situation for migration decisions.
11. In addition, to test for robustness, we used the regionwide ratio of nonfunctioning enterprises to the total number of enterprises as an alternative instrument for remittances. These results are not reported but are available from the authors upon request.
12. Interaction of regional instruments with household characteristics was used, among others, by Hanson and Woodruff (2003) and Amuedo-Dorantes and Pozo (2006).
13. In the survey, respondents were asked to evaluate the conditions of their primary residence by choosing between excellent, good, normal, and not good. As a result, three dummy variables were included in equations (1) to (3), with "excellent" being the omitted category.
14. The J-test, however, suggests that the instruments (i.e. unemployment among men and the share of the vulnerable population) are correlated with the *Banking* variable. It is likely that economic and social development determine the number/availability of commercial bank branches in the regions, thus influencing households' ability to borrow from the banking sector. Given this correlation between the instruments and *Banking*, the results in column (5) of Table 2 should be treated with caution.
15. We owe this insight to Vahan Grigoryan, DDS.
16. The latter result could either mean that women pay more attention to the education-related needs of children or that female students are more expensive than male students (as they are likely to attend more extracurricular activities, such as music, arts, etc.).
17. This is also consistent with the coefficient on the size of the household, where smaller households borrow less.